

Module 3: Continued Fraction Obstruction for $\pi/10$

Certifies: $a_6 = 733$, $Q_5 = 226$, $a_7 = 11$, bound = 82,829

Machine Certificate v1.6 David Fox May 21, 2026

STATUS: CERTIFIED ✓

Source and stdout SHA-256 verified.

Symbol disambiguation (Battle Plan v1.6): Q_n = denominator q_n of the n th convergent of $\pi/10$ (NOT $6^4 = 1296$). a_n = n th partial quotient of $\pi/10$. Values $Q_5 = 1296$ and $a_7 = 1$ are deprecated and shall not appear.

1. Claim

The simple continued fraction of $\pi/10$ is

$$\pi/10 = [0; 3, 5, 2, 5, 1, 733, 11, 1, 1, \dots]$$

with convergents p_n/q_n . In particular, $a_6 = 733$, $q_5 = 226$, $a_7 = 11$. By Lemma 3.2 of Paper 6, for $n \geq 5$ the numerators satisfy $p_n > a_{n+1} \cdot q_n/2$. Hence for $n = 5$:

$$p_5 > 733 \times 226/2 = 82,829.$$

2. Certified Values

Value	Description
$Q_5 = 226$	5th convergent denominator of $\pi/10$
$a_6 = 733$	6th partial quotient of $\pi/10$
$a_7 = 11$	7th partial quotient of $\pi/10$
82,829	Exact integer arithmetic: $733 \times 226/2$

Values $Q_5 = 1296$ and $a_7 = 1$ are false and shall not appear in any certificate.

3. Source Code

File: cf_pi10.py

```
# Battle Plan v1.6 - Module 3: CF of pi/10
# Library: mpmath 1.3.0 (pinned for Modules 1-3)
# Seed fix: p=1,pp=0 tracks numerators; q=0,qq=1 tracks denominators
from mpmath import mp
mp.dps = 50

x = mp.pi/10
a = []
for _ in range(8):
    ai = int(x)
    a.append(ai)
```

```

    x = x - ai
    if x == 0: break
    x = 1/x

# a = [0, 3, 5, 2, 5, 1, 733, 11]
p, q = 1, 0
pp, qq = 0, 1
for i, ai in enumerate(a):
    p, pp = ai*p + pp, p
    q, qq = ai*q + qq, q
    if i == 5: # n=5
        p5, q5 = p, q # 71, 226

a6 = a[6] # 733
a7 = a[7] # 11
bound = a6 * q5 // 2 # 82829

print(f"CF: {a}")
print(f"p_5={p5}, Q_5={q5}, a_6={a6}, a_7={a7}, bound={bound}")

```

Seed fix note: The original LaTeX draft had seeds $p=0, pp=1, q=1, qq=0$, which swaps numerators and denominators, giving $p5=226$ (wrong) and $q5=71$ (wrong). Fixed to $p=1, pp=0, q=0, qq=1$.

4. Build Environment

```

$ python3 --version
Python 3.10.12

$ pip show mpmath | grep Version
Version: 1.3.0

$ sha256sum cf_pi10.py
5ac750a4013027495d76ddd22fc76842f408ef716ba1498437d29414da8169b2
cf_pi10.py

```

5. Raw Execution Log

```

$ python3 cf_pi10.py
CF: [0, 3, 5, 2, 5, 1, 733, 11]
p_5=71, Q_5=226, a_6=733, a_7=11, bound=82829

```

6. Cryptographic Binding

Item	SHA-256 Digest
Source cf_pi10.py	5ac750a4013027495d76ddd22fc76842f408ef716ba1498437d29414da8169b2
Stdout (2 lines)	e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044

7. Causal Dependency

Module 4 requires the inequality $p_5 > 82,829$ to prove that the enumeration of $S(\alpha_0)$ is complete for $p \leq 10^{4000}$. If SHA `e687bb09...` changes, Module 4 must be re-verified. The bound 474,984 is deprecated and invalid.

Module 3 has *no* causal parent SHA — it depends only on the mathematical constant π (computed via `mpmath` 1.3.0) and pure integer arithmetic.

Verification command:

```
$ python3 cf_pi10.py | sha256sum
e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044  -
```

Repository: <https://github.com/DavidFox998/alpha0-ponti> · Certificate generated by Machine Certificate v1.6